

11 — Preference Catalog Schema

Status: DRAFT v0.2 (2026-06-12, rev 2026-06-29)

One place that defines every preference — a query parameter the system may need to ask or assume.

Cards only pick a default for each preference. This file says what the preference is and what each answer changes.

Purpose

A preference is one query parameter the system needs resolved before it can run, like "Primary or Secondary sales?".

The catalog answers, for each preference:

- What is the question?
- What are the options?
- What does each option change? (which metric, unit, baseline, period, or filter)

We define each preference **once** here. It is industry-generic, so it is the same for every buyer and every role. The only per-role part is the default, which lives in the card.

This is the idea from our chat: **map everything, default most, ask few.**

The "map everything" part is this file.

Phase 1 behaviour (decided 2026-06-12)

- We show the options with a default.
- The user confirms at first login.
- The user can change a preference any time. The active session can override it.
- We do NOT learn habits yet (User Model layers 3 + 4 come in a later phase).

Design rules

- One preference = one entry = one id.

- The preference lives here. The default for a role lives in the persona card.
- Each option says what it maps to. That is how an answer becomes a real query change.
- A preference is always reversible. The user can change it later.

Field table

Field	Type	Required	Notes
preference_id	string (snake_case)	yes	stable id, e.g. sales_type
category	enum (see below)	yes	groups preferences for the UI
question_text	string	yes	the question to ask if the preference is unknown
options	Option[]	yes	the choices
options[].value	string	yes	stable value id, e.g. primary
options[].label	string	yes	text shown to the user
options[].maps_to	object	yes	what choosing this option sets (see below)
reversible	boolean	yes	always true for now

category values: data_source , measure , baseline , time_framing , trend , comparison , granularity , channel , brand , output_style .

maps_to keys (all optional; use the ones that fit the preference):

Key	Means	Allowed values
measure	use this metric id as the main measure	a metric id from 12
unit	how to count	pcs_packs volume value
comparison_metric	which metric id to compare against	a metric id from 12

Key	Means	Allowed values
comparison_period	which period to compare against	previous_month previous_quarter smly
baseline_window	history window for averages/anomaly	13m 16m 113m
trend_grain	bucket size for trend charts	day week month
dimension	which dimension to drill into	a Cube.js dimension
filter	a filter to apply	object, e.g. {period: MTD}

JSON Schema

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://cpg-analytics/spec/catalog.schema.json",
  "title": "PreferenceCatalog",
  "type": "array",
  "items": {
    "type": "object",
    "additionalProperties": false,
    "required": ["preference_id", "category", "question_text", "options", "reversible"],
    "properties": {
      "preference_id": { "type": "string", "pattern": "^[a-z][a-z0-9_]*$" },
      "category": { "enum": ["data_source", "measure", "baseline", "time_framing",
                            "trend", "comparison", "granularity", "channel",
                            "brand", "output_style"] },
      "question_text": { "type": "string", "minLength": 1 },
      "reversible": { "type": "boolean" },
      "options": {
        "type": "array",
        "minItems": 2,
        "items": {
          "type": "object",
          "additionalProperties": false,
          "required": ["value", "label", "maps_to"],
          "properties": {
            "value": { "type": "string", "pattern": "^[a-z][a-z0-9_]*$" },
            "label": { "type": "string", "minLength": 1 },
            "maps_to": {
              "type": "object",
              "additionalProperties": false,
              "properties": {
                "measure": { "type": "string" },
                "unit": { "enum": ["pcs_packs", "volume", "value"] },
                "comparison_metric": { "type": "string" },
                "comparison_period": { "enum": ["previous_month", "previous_quarter", "smly"] },
                "baseline_window": { "enum": ["13m", "16m", "113m"] },
                "trend_grain": { "enum": ["day", "week", "month"] },
                "dimension": { "type": "string" },
                "filter": { "type": "object" }
              }
            }
          }
        }
      }
    }
  }
}
```

```

    }
  }
}
}
}
}

```

Validation rules (check in CI)

1. Every `preference_id` is unique.
2. Every `options[].value` is unique inside its preference.
3. Any `measure` or `comparison_metric` in `maps_to` must exist in the metric registry (12).
4. Any `dimension` in `maps_to` must be a valid Cube.js dimension.
5. A card's `preferences[].preference` must match an existing `preference_id` in this catalog (checked in 10).

Versioning

- Add a new preference or option → fine, no break.
- Remove a preference/option, or change a `value` → breaking. Cards using it must be updated.

Worked example

See `examples/catalog.yaml` . One preference:

```
- preference_id: sales_type
category: data_source
question_text: "Which sales view do you usually work with?"
reversible: true
options:
  - value: secondary
    label: "Secondary"
    maps_to: { measure: secondary_net_sales }
  - value: primary
    label: "Primary"
    maps_to: { measure: net_sales }
```